Yasin Rezvani

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Education

Shahrood University of Technology — Shahrood, Semnan Province, Iran
B.Sc. in Computer Engineering — *Sep 2022 – Sep 2027*

- GPA: **3.32**/4.00, 16.73/20
- Major GPA: **3.67**/4.00, 17.67/20
- Thesis: "Brain Tumor MRI Dataset and Joint Deep Neural Network for Segmentation and Classification"
- Advisor: Dr. Mansoor Fateh

Research Interests

Deep Learning for Healthcare, Medical Image Segmentation and Classification, Vision Transformers, Lightweight Neural Networks

Publications

- A. Fateh, **Y. Rezvani**, S. Rezvani, M. Fateh, V. Abolghasemi, "Swin-HAFNet: A Hierarchical Multi-Task Transformer for Brain Tumor Segmentation and Classification," *npj Precision Oncology*, under review, Feb. 2026.
- A. Fateh, **Y. Rezvani**, S. Moayedi, S. Rezvani, F. Fateh, M. Fateh, V. Abolghasemi, "BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification," *Scientific Data*, Feb. 2026.
- S. Rezvani, F. S. Siahkar, **Y. Rezvani**, A. A. Gharahbagh, V. Abolghasemi, "Single Image Denoising via a New Lightweight Learning-Based Model," *IEEE Access*, Aug. 2024.
- **Y. Rezvani**, A. Rezvani, "The Serious Water Crisis and Our Responsibility Against It," *Third National Conference on Modern Approaches to Education and Research*, Mar. 2019.

Research Experience

CVLab SHUT — Shahrood, Iran
Undergraduate Research Assistant — *Jan 2024 – Present*

- Worked on the BRISC 2025 dataset and co-developed the Swin-HAFNet model for brain tumor segmentation and classification, as well as a lightweight image denoising project. Gained hands-on experience in deep learning pipelines, experimental design, and scientific writing.

Shahaab Co. — Shahrood, Iran
Computer Vision Research Intern — *Jun 2024 – Sep 2024*

- Developed YOLO-based models for Persian license plate recognition and Iranian car type classification. Involved in data collection, labeling, and model optimization, gaining practical experience with computer vision libraries and industry applications.

Academic Service

Peer Reviewer
Scientific Data — *Sep 2025 – Present*

- Reviewed a manuscript focused on scientific dataset creation, data collection, annotation, and preprocessing.

Peer Reviewer
IEEE Access — *Jun 2024 – Present*

- Reviewed 14 manuscripts in computer vision, medical image analysis, and image processing.

Speaker
TED-Ed Student Talks — *Jun 2019 – Sep 2019*

- Presented *U-Health*, an AI-driven platform to improve remote access to medical services for all communities.

Projects

LL(1) Parser with Panic Mode Recovery

Supervisor: Dr. Elias Khajeh Karimi — Course: Principles of Compiler Design *Feb 2025 – Jun 2025*

- An interactive parser that visualizes parse trees in real-time and offers step-by-step error recovery.

Binary Search Tree (BST) Visualizer

Supervisor: Dr. Maryam Khodabakhsh — Course: Data Structure *Oct 2024 – Jan 2025*

- A Java Swing-based tool that visualizes BST insert and delete operations with zoom and pan capabilities.

Image Denoising Using FFT and DnCNN

Supervisor: Dr. Esmaeel Tahanian — Course: Engineering Mathematics *Feb 2024 – Jun 2024*

- Image denoising using traditional FFT-based filtering and deep learning (DnCNN) for comparative performance analysis.

Hands-On Machine Learning and Deep Learning with Python

Supervisor: Dr. Mansoor Fateh — Organization: CVLab SHUT *Jan 2024 – Apr 2024*

- A practical collection of Jupyter notebooks covering deep learning, computer vision, and data analysis with PyTorch, TensorFlow, and NumPy/Pandas.

Skills

Programming Languages: Python, C++, Java

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, YOLO, NumPy, Pandas, Matplotlib, SciPy

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab (GPU/TPU), Docker

Certificates

- **Research Collaborations Module** – Elsevier, Jun 2025
- **Prompt Engineering for Everyone** – Cognitive Class, Feb 2024
- **Supervised Machine Learning: Regression and Classification** – Coursera, Feb 2024
- **Machine Learning with Python** – Cognitive Class, Feb 2024
- **Learning Python** – LinkedIn, Aug 2022

References

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